

A Leakage-Safe Benchmark for Multimodal Auditory Attention Decoding

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I. BENCHMARK

A. Motivation and Scope

A dataset is only as useful as the evidence that it contains the signal it claims to. For an auditory attention decoding (AAD) corpus this is subtle: the electroencephalographic (EEG) correlate of selective attention is real but faint, and the richness that makes a multimodal corpus valuable—simultaneous EEG, eye tracking, egocentric scene video, head inertial measurement (IMU), and spatialized multi-speaker audio—also multiplies the ways a decoder can appear to work while decoding something other than the listener’s neural state.

This benchmark makes three contributions. First, a *methodological* one that is its organizing thesis: on this kind of corpus, *content-disjoint splits are necessary but not sufficient* for an admissible neural claim, and must be paired with an *EEG-shuffle null*. We show that a model with a learned stimulus encoder can decode a stimulus-design confound (the attended talker is acoustically marked) even on *unseen* content, so no split removes it—only a null that scrambles the EEG relative to the trial exposes whether a model’s accuracy depends on the brain at all. Second, we *adjudicate* a public benchmark model under this protocol and find its accuracy to be entirely non-neural, quantifying how much of a headline number a split-only protocol can miss. Third, we provide a *truthful baseline*: an EEG-only stimulus-reconstruction decoder whose accuracy is significantly above its own null, together with reference floors, the confound characterization, and the decision-latency trade-off future methods should report.

We restrict the headline decoder to EEG, and report eye tracking, video, and IMU only as *characterization* branches (Section I-F); fusing them is reserved for a companion method paper.

B. Task and Chance Level

Each trial presents four attendable talkers from fixed loudspeakers (plus two distractor loudspeakers that are never attended); the listener attends one and is tested for comprehension. The benchmark task is the four-way identification of the attended talker, evaluated as a *match–mismatch* decision. From EEG we reconstruct a feature $\hat{\mathbf{r}} \in \mathbb{R}^{d \times T}$ of the attended speech and score each of the four real, co-present talker envelopes $\mathbf{c}_k$ by their correlation,

$$s_k = \frac{1}{d} \sum_{b=1}^d \tanh^{-1}(\rho(\hat{\mathbf{r}}_b, \tilde{\mathbf{c}}_b^k)), \quad \hat{y} = \arg \max_k s_k, \quad (1)$$

where ρ is the Pearson correlation over time, $\tilde{\mathbf{c}}$ is the per-band z -scored candidate, and the Fisher z -transform is applied

before the band average. Because the four candidates are the four *real* talkers present in every window, the number of choices is exactly four at any window length; the theoretical chance level is thus 0.25 at every decision window, from 5 s to the whole ≈ 30 s trial. This is the property a four-way baseline must have, and the reason we decode over the real co-present talkers rather than time-shifted surrogates, whose chance level drifts with window length.

Two corpus confounds are defused *architecturally*. *Loudness*: the attended talker is mixed to be attendable, but (1) is scale-free, so an uninformative reconstruction correlates ≈ 0 with every candidate and amplitude cannot help. *Clip identity and the deterministic schedule*: the same stimuli replay to every subject and the attended talker follows a fixed rotation, so we *permute* the four candidate streams per (subject, trial) and predict the attended *slot*; the same stimulus lands on different slots for different subjects, and the decoder has no path from audio to the label except through the EEG-derived reconstruction in (1).

C. Admissibility: Two Necessary Controls

We call a four-way result *admissible* only if it passes both of the following.

(C1) Content-disjoint splits. Because the same stimuli are replayed to every subject, holding out a subject does not hold out the audio. We therefore split by trial content: a `trial_k` in test appears in neither train nor validation. *within*-subject uses a per-subject 5-fold over trial content; *leave-one-subject-out* (LOSO) holds out the subject *and* the content (test = the held-out subject’s held-out trials; train/validation = the other fifteen subjects’ remaining content). Streams are read from a single pre-aligned representation (per-stream recording lags resolved once, audio resampled to a common 64 Hz grid), so cross-modal misalignment cannot be reintroduced, and models are BatchNorm-free so no train-subject statistic leaks under LOSO.

(C2) EEG-shuffle null. We permute the test EEG across trials so the brain data no longer matches the trial’s candidates (same EEG, candidates, and labels; averaged over shuffles). A decoder that uses the EEG collapses to chance; a decoder that ignores the EEG is unchanged. *The null is a property of the model, not the data*: two models on identical data can have very different nulls because the null measures which input a model routes its decision through. A model is admissible only if its accuracy is significantly above *its own* null (paired one-sided t).

Why C1 alone is insufficient. The attended talker is systematically enhanced by the stimulus design, leaving a

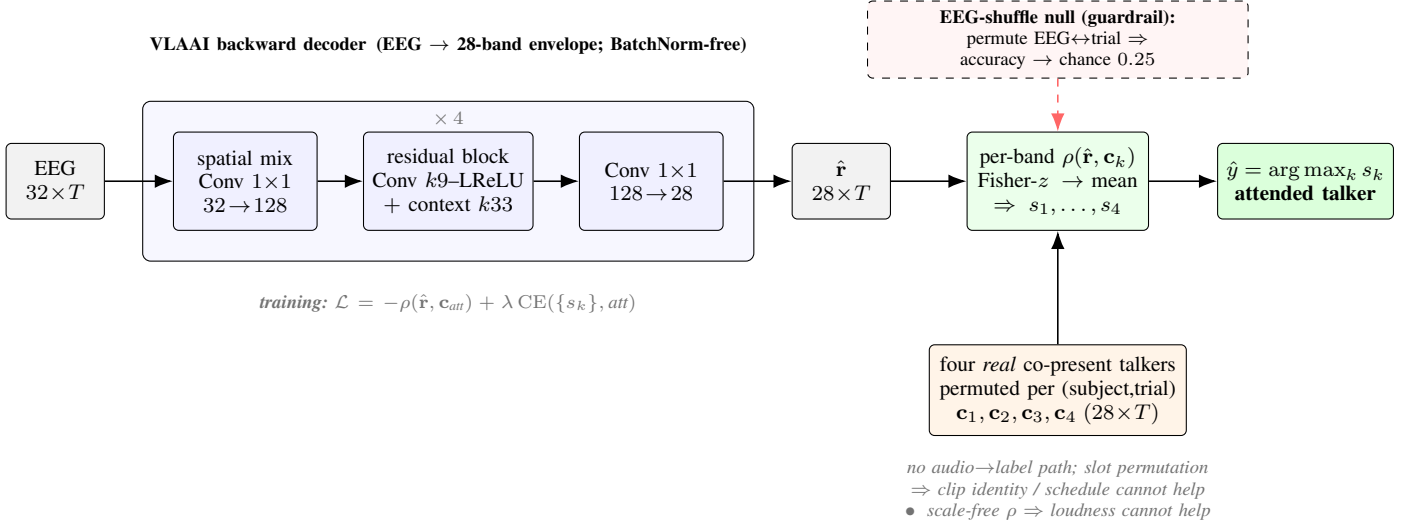


Fig. 1. **The benchmark decoder and its admissibility controls.** A VLAAI-style backward network reconstructs the attended talker’s 28-band envelope $\hat{\mathbf{r}}$ from 32-channel EEG—a 1×1 spatial mix, four residual convolutional blocks with output-context temporal mixing, and a 1×1 projection, with no batch normalization. The four *real* co-present talker envelopes, permuted per (subject, trial), are scored by their per-band Fisher- z correlation with $\hat{\mathbf{r}}$ (1); $\arg \max$ gives the attended talker (chance 0.25 at every window). The scale-free correlation defuses loudness and the slot permutation defuses clip identity and the deterministic schedule; because the decoder has *no audio-to-label path*, the EEG-shuffle null—permuting the EEG against the trial—drops accuracy to chance, certifying the decision is neural (Section I-C). A covert-spatial branch (gaze-residualized posterior α/β power $\rightarrow$ shrinkage LDA) is fused out-of-fold within-subject; gaze, video, and IMU are reported only as orienting-characterization branches.

general acoustic marking present in every trial, unseen ones included. A model with a learned *audio* encoder can detect this “attended-style” envelope on any track, so its accuracy survives content-disjoint splitting—the confound is a stimulus-design property, not track memorization, and no split can remove it. Only C2 reveals whether a given model exploits it. A causal-lag control (accuracy at positive vs. negative EEG-to-audio lag; genuine cortical tracking is causal, with audio leading EEG by ≈ 100 –250 ms) is logged as a secondary check.

D. Models

Content decoder (headline). A *backward* (stimulus-reconstruction) network maps EEG to a speech feature and decides by (1) (Fig. 1). The headline model is VLAAI-style [4]: a learned $32 \rightarrow 128$ spatial projection, four residual convolutional blocks with output-context temporal mixing, and a 1×1 projection to a 28-band mel-spectrogram target; *no* batch normalization. Two confound-free additions are used: *multi-band* reconstruction (fuse per-band correlations via the Fisher z -transform of (1)) and a *match-mismatch margin* (an auxiliary cross-entropy over the four candidate correlations). Accuracies are 5-seed reconstruction ensembles. A fully-convolutional decoder trained at 5 s scores at any window, so we report a *decision-window curve*.

Reference floor. A single causal Conv1d integrating a ≈ 1 s lag across channels—the classical linear AAD decoder [1], [2], [3]. It is the field-canonical floor and, as shown below, sits at chance, certifying the task admits no trivial linear leak.

Adjudicated prior. We port the public benchmark’s classifier [5]: dilated-convolutional encoders for EEG *and* for each candidate audio into a common space, with a learned cosine-similarity head and a softmax over candidates. Its *learned*

audio encoder is precisely the “audio-to-label path” that C2 is designed to expose.

Covert-spatial branch. Log α/β band-power over posterior channels, *gaze-residualized* (the window’s gaze is regressed out with a train-fold fit) and classified by a shrinkage LDA [7]; it reads covert neural direction, not eye movements. It is fused with the content decoder by a late weight tuned strictly out-of-fold.

Overt-orienting characterization branches. Two-dimensional gaze, frozen scene+fovea video embeddings (a self-supervised encoder [8]), and a head-IMU summary, each decoded for attended *direction* and attended *content*. These delimit the confounds and are not part of the headline decoder.

E. Training

Content decoders reconstruct the attended feature under a negative-Pearson loss with AdamW, cosine schedule with warmup, gradient clipping, and early stopping on inner-validation match-mismatch accuracy; the VLAAI model uses learning rate 10^{-3} , weight decay 10^{-4} , batch 128, dropout 0.2, up to 60–80 epochs, and margin weight 0.3. Each seed’s reconstruction is z -scored over time and averaged before (1). Training windows are 5 s with 50% overlap; the spatial LDA and overt classifiers are refit per evaluated window. The adjudicated prior is trained with its published recipe (Adam, label smoothing, early stopping on inner-validation) but under C1/C2.

F. Results

Accuracies are means over 16 subjects, held-out test, four-way (chance 0.25); “null” is the EEG-shuffle null and “margin” is accuracy minus null.

TABLE I

FOUR-WAY ACCURACY AT 5 S UNDER CONTENT-DISJOINT SPLITS (C1), WITH THE EEG-SHUFFLE NULL (C2). MARGIN SIGNIFICANCE IS A PAIRED ONE-SIDED t ($n=16$).

Model	Protocol	4-way	null	margin (p)
Ours (VLA AI, mb+margin)	LOSO	0.34	0.26	+0.07 (2×10^{-3})
Ours	within	0.30	0.26	+0.045 (7×10^{-3})
Linear (reference floor)	both	0.25	0.25	≈ 0 (n.s.)
Adjudicated prior [5]	LOSO	0.362	0.362	-0.00 (n.s.)
Adjudicated prior	within	0.253	0.253	-0.00 (n.s.)

TABLE II

DECISION-WINDOW CURVE (16 SUBJECTS, VLA AI MB+MARGIN). Δ IS THE PAIRED MARGIN OVER THE EEG-SHUFFLE NULL; ALL Δ SIGNIFICANT AT $p < 10^{-10}$.

Window	LOSO			within		
	4-way	null	Δ	4-way	null	Δ
5 s	0.348	0.266	+0.083	0.317	0.259	+0.058
10 s	0.383	0.264	+0.119	0.336	0.261	+0.075
15 s	0.431	0.264	+0.167	0.354	0.257	+0.097
20 s	0.440	0.262	+0.178	0.343	0.256	+0.087
30 s	0.497	0.267	+0.230	0.406	0.261	+0.145

Admissibility, side by side. Table I reports the 5 s point under C1. Our decoder is significantly above its null in both protocols; the linear decoder is at chance (no trivial leak); and the adjudicated prior’s accuracy *equals its own null*—content-disjoint splitting does not lower it, because its learned audio encoder reads the acoustic marking on unseen tracks. Its higher raw number is therefore not a better read of the brain: none of it is neural.

Headline: decision-window curve. Accuracy integrates over time (Table II). The four-way rate rises with the window while the null stays flat at ≈ 0.26 , so the neural margin grows and stays highly significant; LOSO reaches 0.497 at the whole trial (margin +0.23, $p < 10^{-13}$) and crosses 0.40 by ≈ 15 s.

What raises the margin, and what does not. We swept improvement levers under C1/C2. The dominant lever is the decision window above. *Multi-band + margin* adds a small significant gain over a plain envelope decoder (LOSO four-way +0.020, $p=0.001$). *K-shot subject calibration*—fine-tuning the LOSO model on K of the held-out subject’s own trials, disjoint from its test trials—adds +0.015 margin at $K=20$ ($p=0.03$), saturating there. Two intuitive levers did *not* survive 16-subject validation: an onset/edge (rectified-derivative) target (within Δ margin -0.026) and distractor hard-negatives (speakers 5–6). Throughout, the null remained at 0.25–0.26: envelope-feature engineering has plateaued, but every reported number is genuinely neural.

Covert-spatial fusion improves the within-subject four-way at every window (e.g. 0.313 $\rightarrow$ 0.347 at 5 s) because the cue is individually calibrated; under LOSO the admissible cross-subject fusion weight goes to zero at every window—the covert-spatial cue is real per subject but does not transfer—so LOSO is carried by content alone. Covert spatial fusion is thus a within-subject lever.

TABLE III

OVERT-ORIENTING REFERENCE BRANCHES: ATTENDED DIRECTION AND CONTENT (WHOLE-TRIAL FOUR-WAY, CHANCE 0.25). “ $\approx$ ch” = AT CHANCE AT EVERY WINDOW.

Modality	Direction		Content	
	within	LOSO	within	LOSO
Gaze (2D)	0.49–0.57	0.42–0.47	$\approx$ ch	$\approx$ ch
Scene video	0.62–0.68	0.55–0.59	$\approx$ ch	$\approx$ ch
Head IMU (accel+gyro)	0.38–0.46	0.31–0.40	$\approx$ ch	$\approx$ ch

Overt-orienting characterization. Table III decodes direction and content from the three non-EEG modalities. Each decodes *direction* well above chance and rising with the window; *none* decodes *content* above chance. Vision is the strongest overt cue, the head IMU the weakest (no magnetometer, so head yaw is only partly recoverable). The unanimity—orienting everywhere, content nowhere—is why the spatial branch is gaze-residualized and the content task permuted.

The null level. Theoretical chance is 0.25 and a white-noise reconstruction scores 0.250 empirically, confirming the four candidates are a fair choice. Our trained decoder’s null instead sits, flatly, at ≈ 0.265 : the 0.015 excess is the *audio-only floor*—a decoder trained to reconstruct attended envelopes matches the attended talker’s marking slightly even with its EEG scrambled. This offset is constant across windows, present equally in the real accuracy, and removed by measuring the margin against the empirical null rather than 0.25; reporting against 0.25 would overstate the effect.

G. Discussion

The central lesson. Strict splits are not enough. A subject- and content-disjoint protocol is standard practice, yet a model with a learned stimulus encoder converted it back into an inadmissible result by reading the stimulus design itself (Table I): its accuracy is its own null. The EEG-shuffle null is the control that catches this, and it is inexpensive—for a window-wise decoder it is exactly a permutation of the reconstruction against the candidates. We recommend it as a required control for any AAD leaderboard, alongside content-disjoint splits and a reported chance level that is the *empirical* null, not $1/K$.

What the benchmark establishes. The corpus carries genuine, confound-free cortical attention signal: under C1/C2 the deep decoder is well above its null (LOSO whole-trial 0.497 vs. 0.267, $p < 10^{-13}$) while the linear floor is at chance. The negative controls are as informative as the positive result—the linear floor rules out a trivial leak, and the overt-modality branches (Table III) show every strong non-neural cue is spatial, not content, so it cannot masquerade as talker decoding once the task is permuted.

Applicability. The decision-window curve quantifies the accuracy–latency trade-off any real-time selection system must navigate [6]: usefully above chance by 5 s, approaching one-in-two at the whole trial. The within-vs-LOSO gap and the K -shot curve quantify the calibration burden: a subject-independent model is carried by content alone, while ≈ 20 calibration trials of a new user add a small further gain and a

subject-specific model can also exploit the covert-spatial cue. The characterization branches tell a designer what each sensor buys—strong, cheap orienting from gaze and video—and warn that none substitutes for the neural content read-out.

Limitations. Envelope tracking here is genuinely weak, so absolute accuracies are modest and envelope-feature engineering has plateaued (onset targets and hard-negatives did not help). With 100 trials per subject the whole-trial estimates are noisy. The empirical null is 0.265 rather than exactly 0.25; we report margins over it. The covert-spatial cue does not transfer across subjects, bounding a subject-independent fusion.

Headroom. By construction the benchmark leaves room above it: the decoder is a single reconstruction with a fixed correlation read-out. A learned similarity head (added to our correlation front-end it is margin-neutral but—unlike the adjudicated prior’s audio encoder—keeps the null at ≈ 0.26 , since it operates on EEG-to-audio correlations rather than raw audio), reliability-weighted multimodal late fusion, subject adaptation, and self-supervised pretraining are all untouched and are the subject of the companion method paper. The benchmark’s role is to fix the task, the two admissibility controls, the windows, and the confound characterization on which those methods should be judged.

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